Week 1 – Virtual Machine & Domain Controller Build

AccessGuard Health – Healthcare IAM Operations Simulation

Hands-on lab record: VirtualBox, Ubuntu Server and a Samba4 Active Directory domain (HVMC.LAN)

Project Name	AccessGuard Health
Project Type	Healthcare IAM Operations Simulation – Home Lab
Organization	Horizon Valley Medical Center (HVMC) – fictional
Project Phase	Lab build: virtualization, domain controller, directory structure
Document Version	1.0
Prepared By	Sabitha S Karanavar
Lab Build Dates	October 25–26, 2026

1. Purpose

This document records the build of the AccessGuard Health home lab that will host the HVMC identity and access management (IAM) simulation. It shows, step by step and with screenshots, how a Windows-style Active Directory domain was created on free software, running on a 2014 MacBook Air. Everything here is a fictional training environment: no real patient data or real credentials are used.

The result is a working domain controller (**hvmc-dc01**) for the domain **HVMC.LAN**, populated with the hospital's 10 departments, 10 role-based security groups and 30 employee accounts, protected by three VM snapshots.

2. Lab Environment

Component	Choice
Host computer	MacBook Air (2014), Intel Core i5, 8 GB RAM, macOS Big Sur 11.7.11
Hypervisor	Oracle VirtualBox 7.0.26
Server OS	Ubuntu Server 24.04.5 LTS
Directory service	Samba 4.19 Active Directory Domain Controller (internal DNS)
Domain / Realm / NetBIOS	hvmc.lan / HVMC.LAN / HVMC
Server name and IP	hvmc-dc01 – 192.168.56.10 (fixed)
Networks	Adapter 1: NAT (internet, 10.0.2.15). Adapter 2: Host-only 192.168.56.0/24 (lab network; Mac = .1)
VM resources	20 GB disk, headless operation, remote access by SSH from macOS Terminal

3. Host Computer and Tool Selection

The 2014 MacBook Air is older hardware, so the lab was kept lightweight: a command-line server with no desktop, which needs far less memory and disk than a graphical one. Free tools were chosen so the project costs nothing.



Figure 1: Host computer: MacBook Air, macOS Big Sur 11.7.11

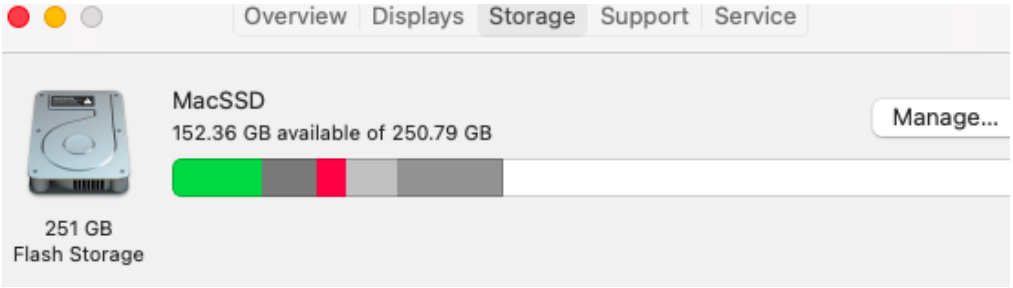


Figure 2: Available storage on the host before the build

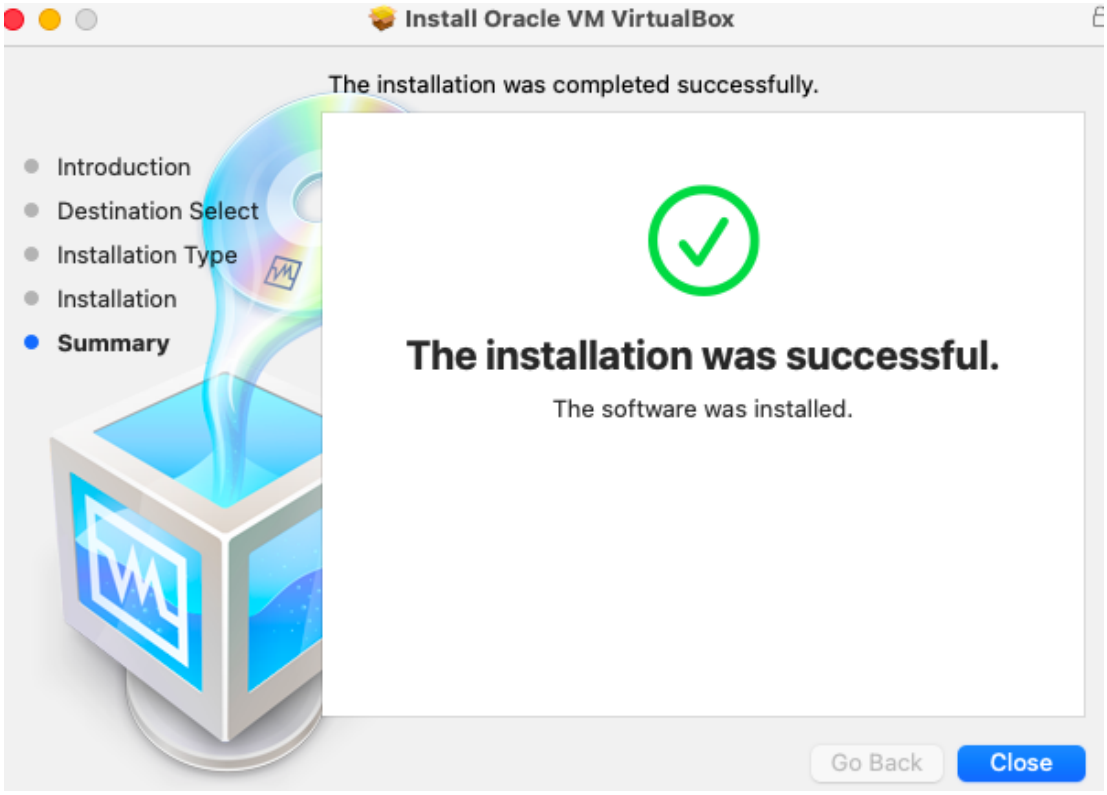


Figure 3: VirtualBox 7.0.26 installed on the Mac

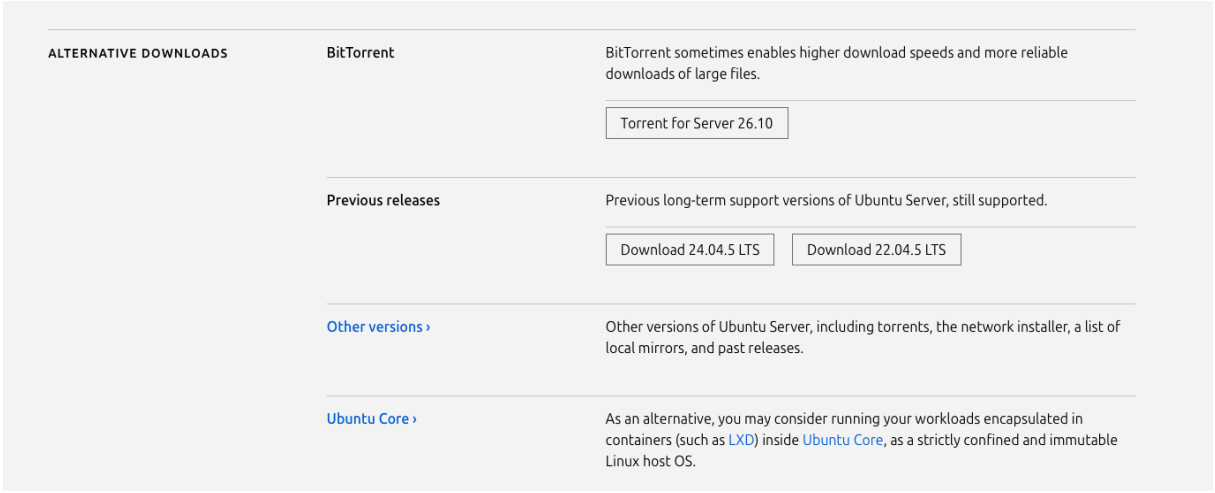


Figure 4: Ubuntu Server 24.04 installer downloaded

4. Virtual Machine Creation

The virtual machine **HVMC-DC01** was created with two network adapters. **Why two?** The NAT adapter gives the server internet access for updates; the host-only adapter creates a private lab network where the server keeps a fixed address and the Mac (and later the client VM) can reach it.

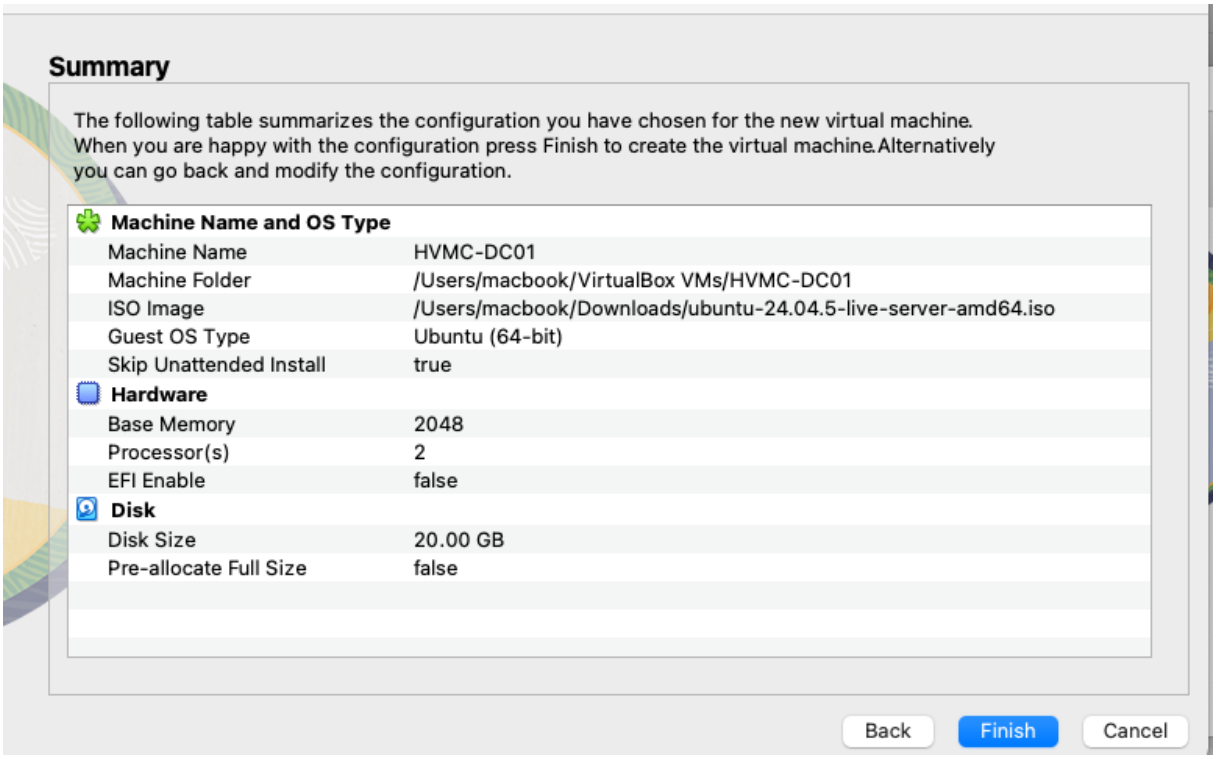


Figure 5: HVMC-DC01 virtual machine summary

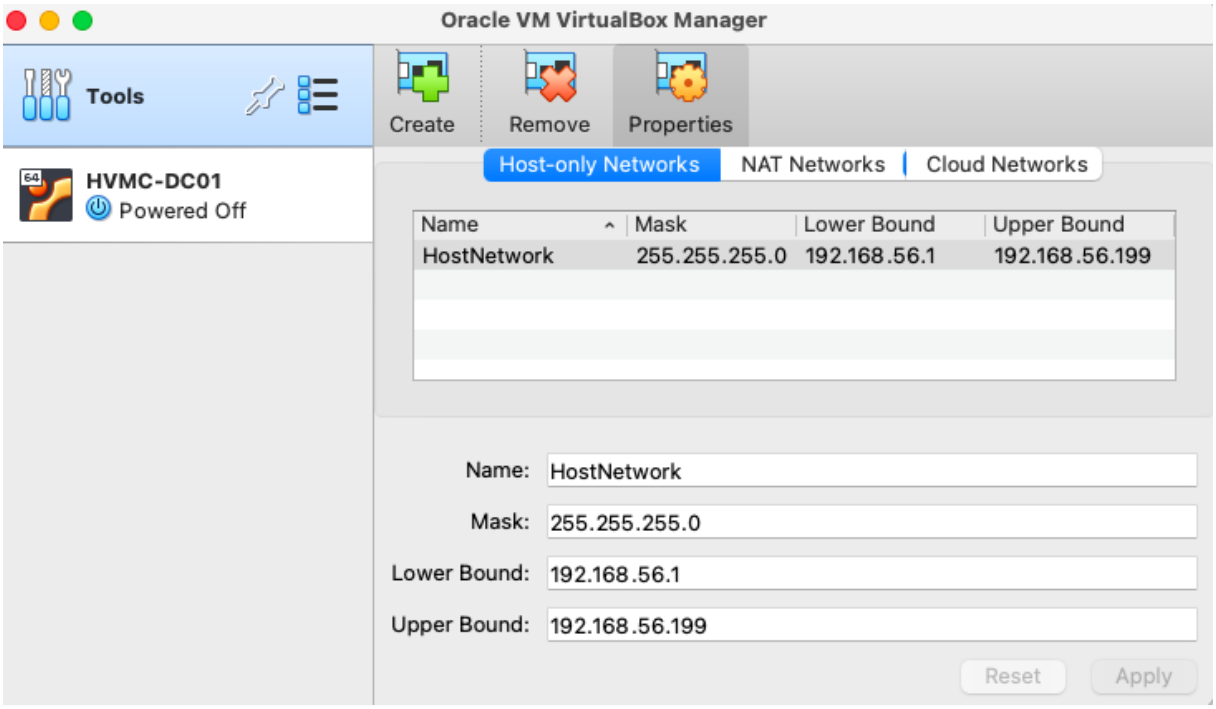


Figure 6: Host-only network 192.168.56.0/24 ("HostNetwork") created in VirtualBox

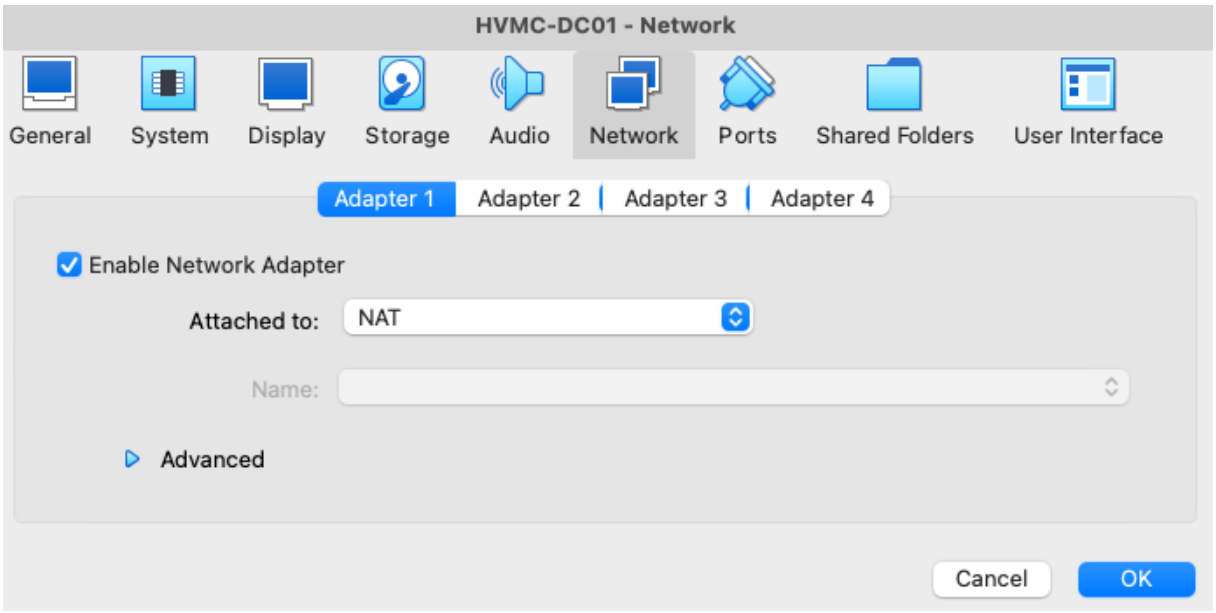


Figure 7: Adapter 1 – NAT

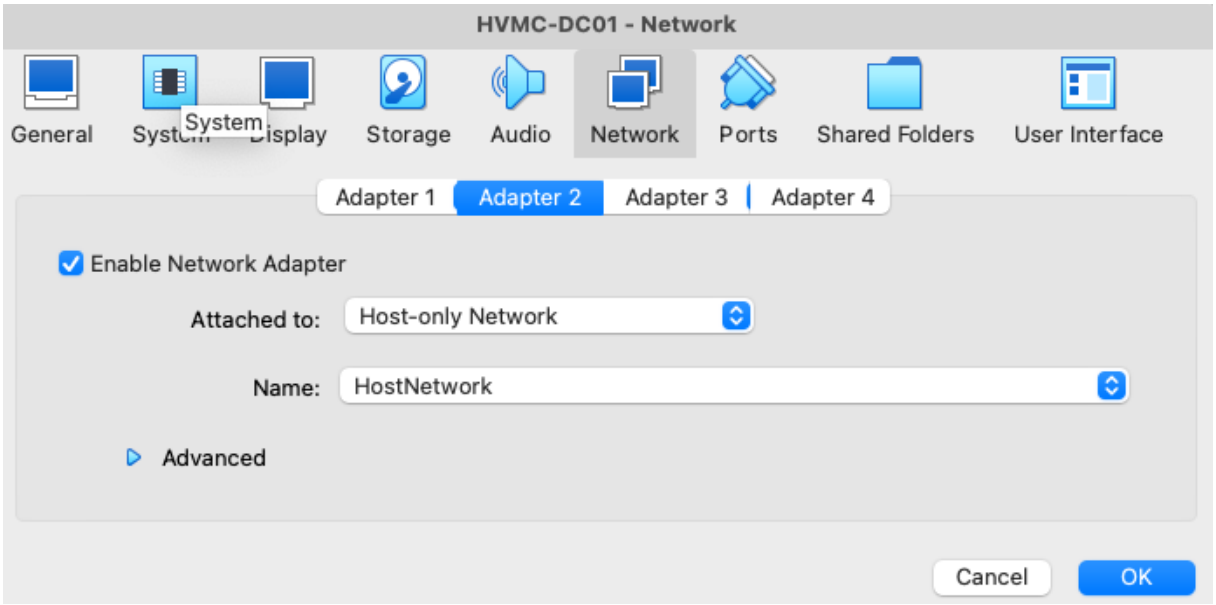


Figure 8: Adapter 2 – Host-only network

5. Ubuntu Server Installation

Setting	Choice	Why
Network	DHCP on both adapters during install	Simple; the fixed IP is set afterwards
Storage	Entire 20 GB disk, LVM	Simplest layout for a lab; LVM allows resizing
Server name	hvmc-dc01	Matches the hospital naming style and becomes the DNS name
OpenSSH server	Installed	Lets us work from the Mac Terminal with copy/paste instead of the small VM window
Extra snaps	None	Keeps the server minimal and secure

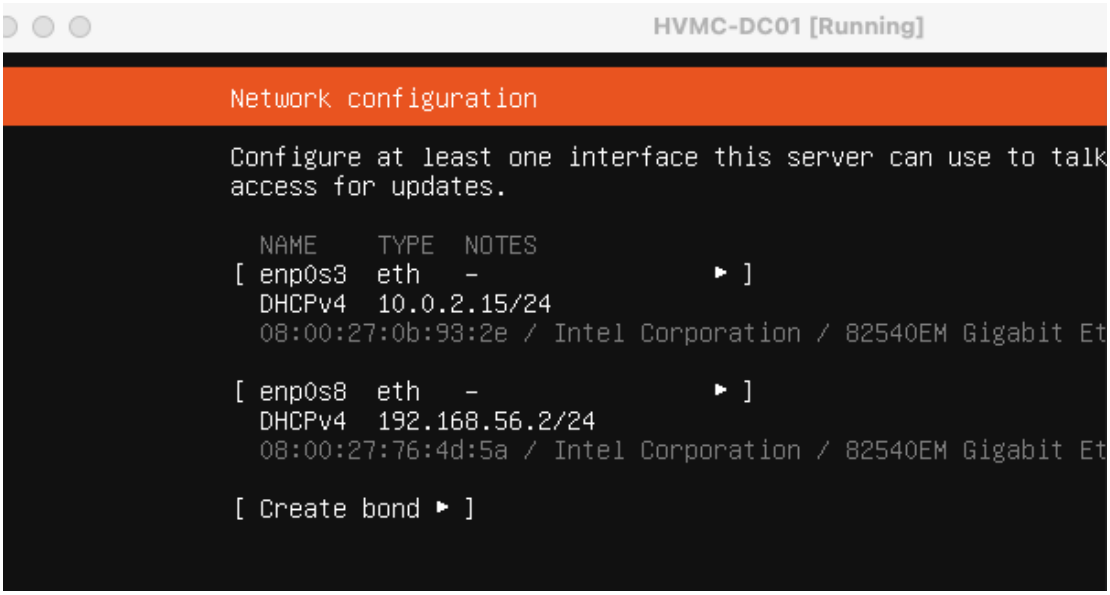


Figure 9: Installer network screen

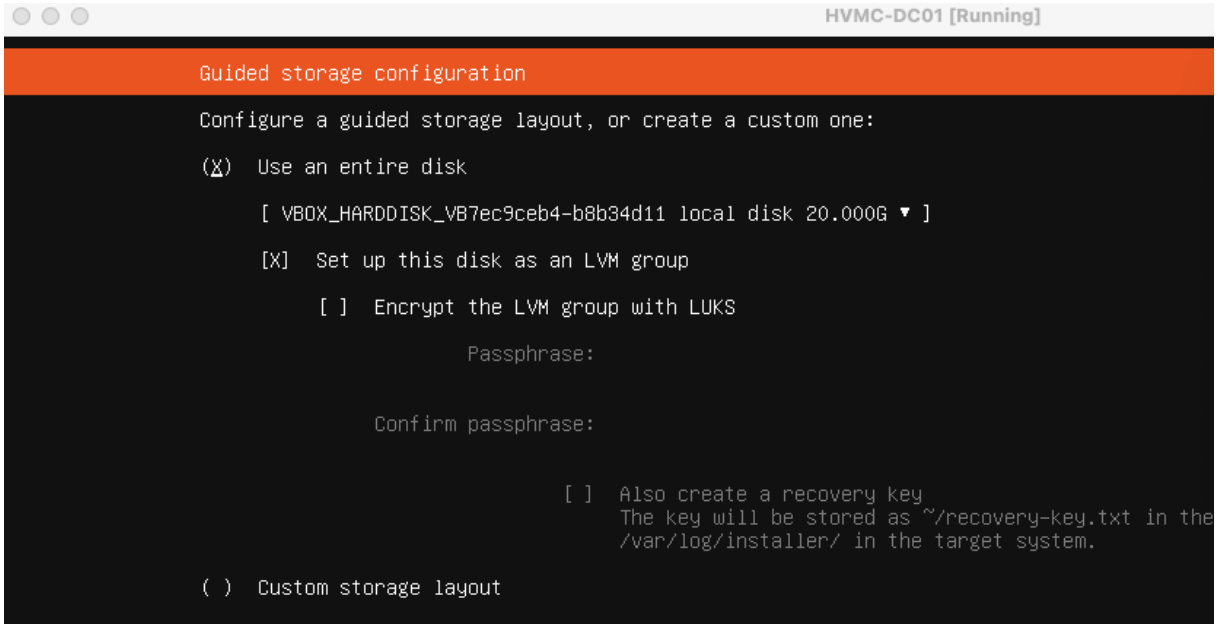


Figure 10: Guided storage configuration (entire disk, LVM)

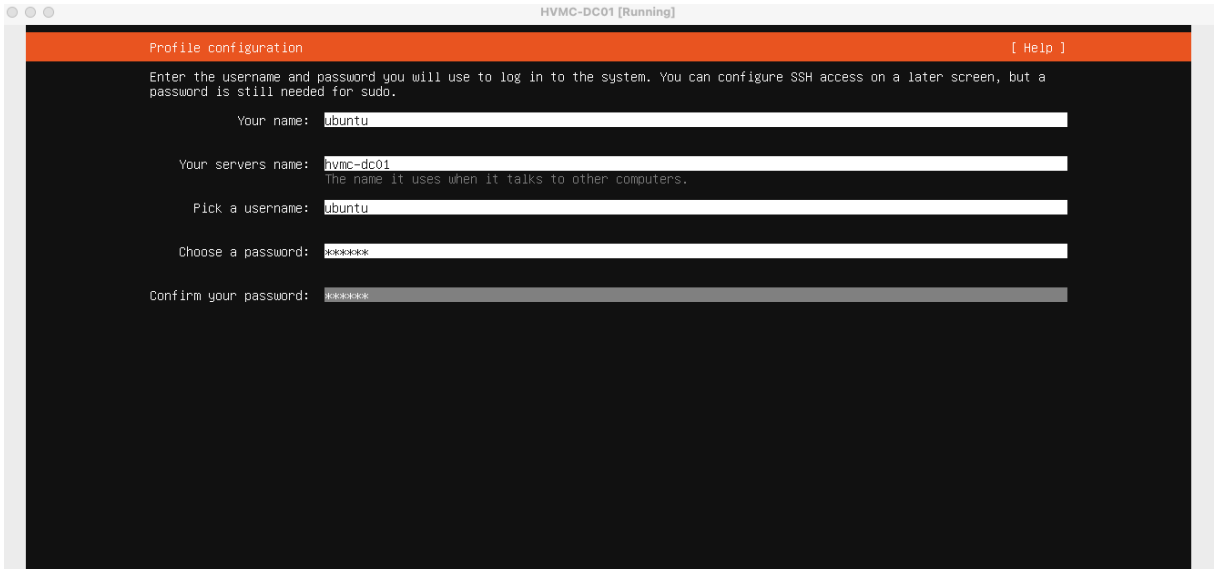


Figure 11: Profile setup: server name and administrative user

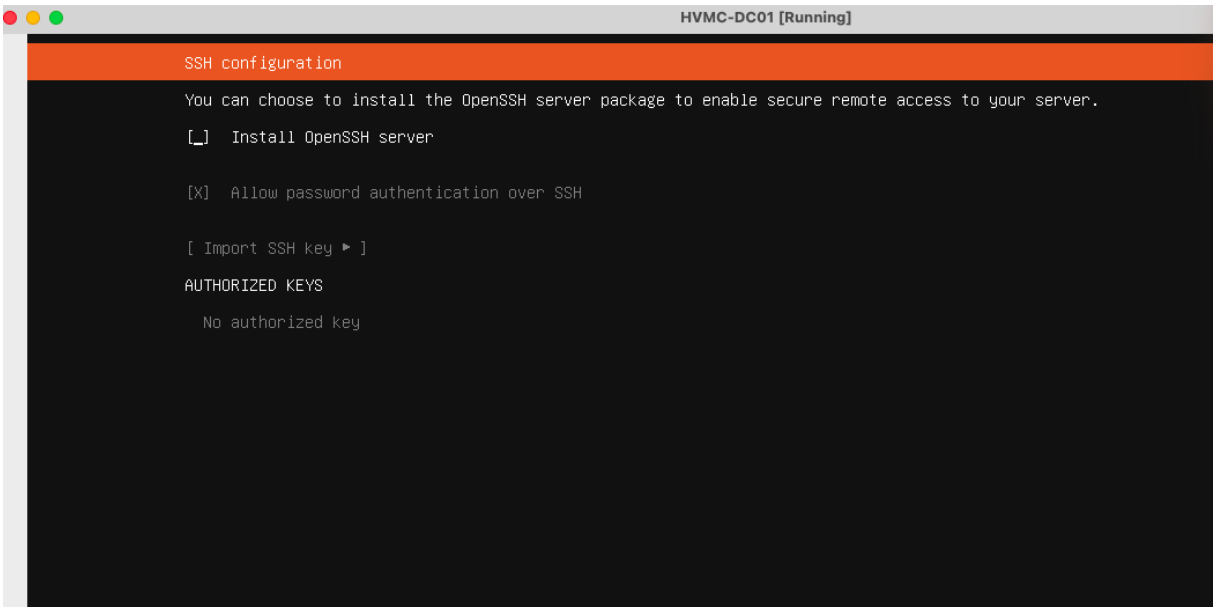


Figure 12: OpenSSH server enabled

6. Network Configuration and Remote Access

A domain controller must never change address, because every other computer finds the domain through it. The server's second adapter (enp0s8) was therefore given a **fixed IP of 192.168.56.10** using a netplan file. Cloud-init's network management was disabled so it cannot overwrite this setting at reboot. After a restart, the server was reached from the Mac over SSH.

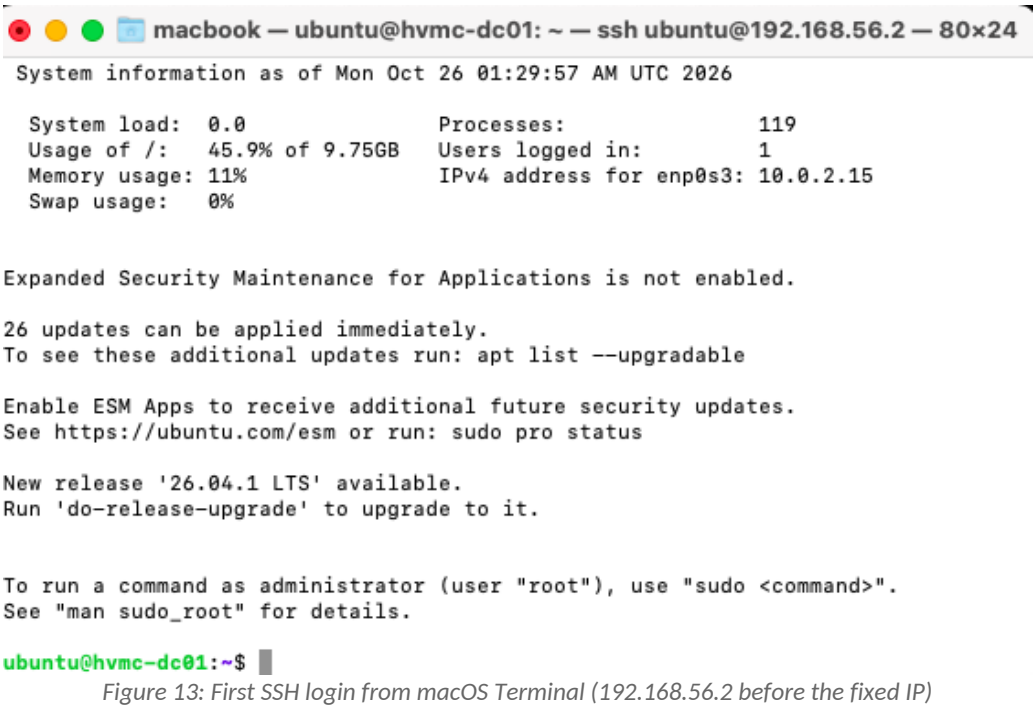


Figure 13: First SSH login from macOS Terminal (192.168.56.2 before the fixed IP)

```
network:
  version: 2
  ethernets:
    enp0s3:
      dhcp4: true
    enp0s8:
      dhcp4: false
      addresses:
        - 192.168.56.10/24
EOF
sudo chmod 600 /etc/netplan/50-cloud-init.yaml
sudo cat /etc/netplan/50-cloud-init.yaml
network:
  version: 2
  ethernets:
    enp0s3:
      dhcp4: true
    enp0s8:
      dhcp4: false
      addresses:
        - 192.168.56.10/24
ubuntu@hvmc-dc01:~$ echo 'network: {config: disabled}' | sudo tee /etc/cloud/cloud.cfg.d/99-disable-network-config.cfg
sudo netplan apply
network: {config: disabled}
client_loop: send disconnect: Broken pipe
Macbooks-MacBook-Pro-2:~ macbook$ ssh ubuntu@192.168.56.10
The authenticity of host '192.168.56.10 (192.168.56.10)' can't be established.
ECDSA key fingerprint is SHA256:+sS1qXaOiS1KnNF7evLBCtNxTHP5+BMp3RCNziWPSww.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.56.10' (ECDSA) to the list of known hosts.
ubuntu@192.168.56.10's password:
Welcome to Ubuntu 24.04.5 LTS (GNU/Linux 6.8.0-147-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Mon Oct 26 01:42:29 AM UTC 2026

  System load:  0.0               Processes:               123
  Usage of /:   45.9% of 9.75GB   Users logged in:         1
  Memory usage: 10%               IPv4 address for enp0s3: 10.0.2.15
  Swap usage:   0%


Expanded Security Maintenance for Applications is not enabled.

26 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

New release '26.04.1 LTS' available.
Run 'do-release-upgrade' to upgrade to it.


Last login: Mon Oct 26 01:29:58 2026 from 192.168.56.1
ubuntu@hvmc-dc01:~$ ip -br a
lo               UNKNOWN        127.0.0.1/8 ::1/128
enp0s3           UP             10.0.2.15/24 metric 100 fe80::a00:27ff:fe0b:932e/64
enp0s8           UP             192.168.56.10/24 fe80::a00:27ff:fe76:4d5a/64
ubuntu@hvmc-dc01:~$
```

Figure 14: Netplan configuration: enp0s3 DHCP (NAT) and enp0s8 static 192.168.56.10

```
macbook — ubuntu@hvmc-dc01: ~ — -bash — 80×24
Service restarts being deferred:
 systemctl restart ModemManager.service
 /etc/needrestart/restart.d/dbus.service
 systemctl restart systemd-logind.service
 systemctl restart unattended-upgrades.service

No containers need to be restarted.

User sessions running outdated binaries:
 ubuntu @ session #1: login[954]
 ubuntu @ session #10: sshd[1545]
 ubuntu @ session #8: sshd[1369]
 ubuntu @ user manager service: systemd[1225]

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@hvmc-dc01:~$ sudo poweroff

Broadcast message from root@hvmc-dc01 on pts/2 (Mon 2026-10-26 01:52:18 UTC):

The system will power off now!

ubuntu@hvmc-dc01:~$ Connection to 192.168.56.10 closed by remote host.
Connection to 192.168.56.10 closed.
Macbooks-MacBook-Pro-2:~ macbook$ 4
```

Figure 15: Safe shutdown with sudo poweroff before taking the first snapshot

```
Last login: Sun Oct 25 21:28:39 on ttys000

The default interactive shell is now zsh.
To update your account to use zsh, please run `chsh -s /bin/zsh`.
For more details, please visit https://support.apple.com/kb/HT208050.
Macbooks-MacBook-Pro-2:~ macbook$ ssh ubuntu@192.168.56.10
ubuntu@192.168.56.10's password:
Welcome to Ubuntu 24.04.5 LTS (GNU/Linux 6.8.0-147-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Mon Oct 26 02:03:05 AM UTC 2026

  System load:  0.17              Processes:               122
  Usage of /:   45.9% of 9.75GB   Users logged in:         0
  Memory usage: 8%                IPv4 address for enp0s3: 10.0.2.15
  Swap usage:   0%

 * Canonical Workshop gives developers fast, composable, reproducible, and
   secure developer environments that are perfect for agentic workflows.

   https://ubuntu.com/workshop

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

New release '26.04.1 LTS' available.
Run 'do-release-upgrade' to upgrade to it.


Last login: Mon Oct 26 01:42:30 2026 from 192.168.56.1
ubuntu@hvmc-dc01:~$
```

Figure 16: SSH login after restart, confirming the fixed address 192.168.56.10

Snapshot 01 – "clean Ubuntu + fixed IP" was taken at this point: a restore point before any directory software is installed.

7. Samba Active Directory Domain Controller

7.1 Installing Samba

Samba, Kerberos and DNS helper packages were installed. Samba is the open-source software that can act as an Active Directory domain controller, speaking the same protocols as Windows Server.

```
Setting up python3-ldb (2:2.8.0+samba4.19.5+dfsg-4ubuntu9.8) ...
Setting up samba-dsdb-modules:amd64 (2:4.19.5+dfsg-4ubuntu9.8) ...
Setting up libsmbclient0:amd64 (2:4.19.5+dfsg-4ubuntu9.8) ...
Setting up libcups2t64:amd64 (2.4.7-1.2ubuntu7.14) ...
Setting up python3-samba (2:4.19.5+dfsg-4ubuntu9.8) ...
Setting up samba-vfs-modules:amd64 (2:4.19.5+dfsg-4ubuntu9.8) ...
Setting up smbclient (2:4.19.5+dfsg-4ubuntu9.8) ...
Setting up samba-common-bin (2:4.19.5+dfsg-4ubuntu9.8) ...
Setting up samba (2:4.19.5+dfsg-4ubuntu9.8) ...
Created symlink /etc/systemd/system/smb.service → /usr/lib/systemd/system/smbd.service.
Created symlink /etc/systemd/system/multi-user.target.wants/smbd.service → /usr/lib/systemd/system/smbd.service.
Created symlink /etc/systemd/system/nmb.service → /usr/lib/systemd/system/nmbd.service.
Created symlink /etc/systemd/system/multi-user.target.wants/nmbd.service → /usr/lib/systemd/system/nmbd.service.
Created symlink /etc/systemd/system/samba.service → /usr/lib/systemd/system/samba-ad-dc.service.
Created symlink /etc/systemd/system/multi-user.target.wants/samba-ad-dc.service → /usr/lib/systemd/system/samba-ad-dc.service.
Setting up winbind (2:4.19.5+dfsg-4ubuntu9.8) ...
Created symlink /etc/systemd/system/multi-user.target.wants/winbind.service → /usr/lib/systemd/system/winbind.service.
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for libc-bin (2.39-0ubuntu8.9) ...
Processing triggers for ufw (0.36.2-6) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@hvmc-dc01:~$
```

Figure 17: Samba and Kerberos packages installed

7.2 Clearing the way and fixing the hostname

Ubuntu installs Samba as a plain file server, which conflicts with the Active Directory role. The file-server services (smbd, nmbd, winbind) were stopped, disabled and masked, and the default smb.conf was moved aside. The hostname was mapped to the fixed IP in /etc/hosts so the server can resolve its own full name (hvmc-dc01.hvmc.lan).

```
ubuntu@hvmc-dc01:~$ sudo systemctl stop smbd nmbd winbind
sudo systemctl disable smbd nmbd winbind
sudo systemctl unmask samba-ad-dc
sudo mv /etc/samba/smb.conf /etc/samba/smb.conf.orig
[sudo] password for ubuntu:
Synchronizing state of smbd.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install disable smbd
Synchronizing state of nmbd.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install disable nmbd
Synchronizing state of winbind.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install disable winbind
Removed "/etc/systemd/system/nmb.service".
Removed "/etc/systemd/system/multi-user.target.wants/smbd.service".
Removed "/etc/systemd/system/multi-user.target.wants/nmbd.service".
Removed "/etc/systemd/system/multi-user.target.wants/winbind.service".
Removed "/etc/systemd/system/smb.service".
ubuntu@hvmc-dc01:~$ sudo sed -i '/hvmc-dc01/d' /etc/hosts
ubuntu@hvmc-dc01:~$ echo "192.168.56.10 hvmc-dc01.hvmc.lan hvmc-dc01" | sudo tee -a /etc/hosts
192.168.56.10 hvmc-dc01.hvmc.lan hvmc-dc01
ubuntu@hvmc-dc01:~$ hostname -f
cat /etc/hosts
hvmc-dc01.hvmc.lan
127.0.0.1 localhost

# The following lines are desirable for IPv6 capable hosts
::1     ip6-localhost ip6-loopback
fe00::0 ip6-localnet
ff00::0 ip6-mcastprefix
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters
192.168.56.10 hvmc-dc01.hvmc.lan hvmc-dc01
ubuntu@hvmc-dc01:~$
```

Figure 18: Services disabled, old configuration moved aside, /etc/hosts updated and checked with hostname -f

7.3 Provisioning the domain

`samba-tool domain provision` built the domain. Key answers: realm **HVMC.LAN**, domain **HVMC**, server role **dc**, DNS backend **SAMBA_INTERNAL** (no extra DNS software to manage), DNS forwarder **8.8.8.8** (so the lab can still look up internet names). The generated Kerberos configuration was copied to /etc/krb5.conf.

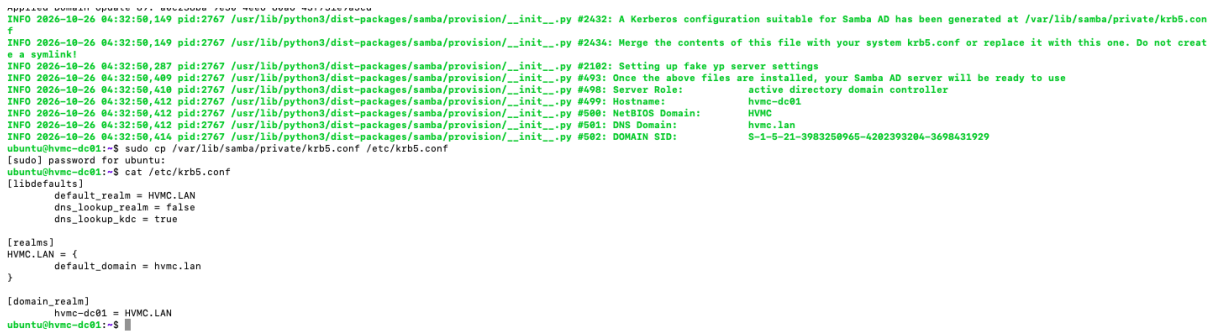


Figure 19: Provisioning completed and krb5.conf installed

7.4 Troubleshooting: DNS port conflict

The Samba service started with red errors: *Failed to bind to 0.0.0.0:53*. Ubuntu's built-in resolver (systemd-resolved) was already using the DNS port. The fix was to disable systemd-resolved, remove the symlinked resolv.conf and write a new one pointing at the domain controller itself (192.168.56.10, search domain hvmc.lan). After restarting samba-ad-dc the service reported **active**.

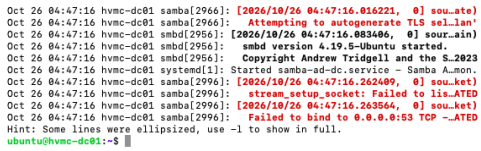


Figure 20: Samba log showing the port 53 bind failure

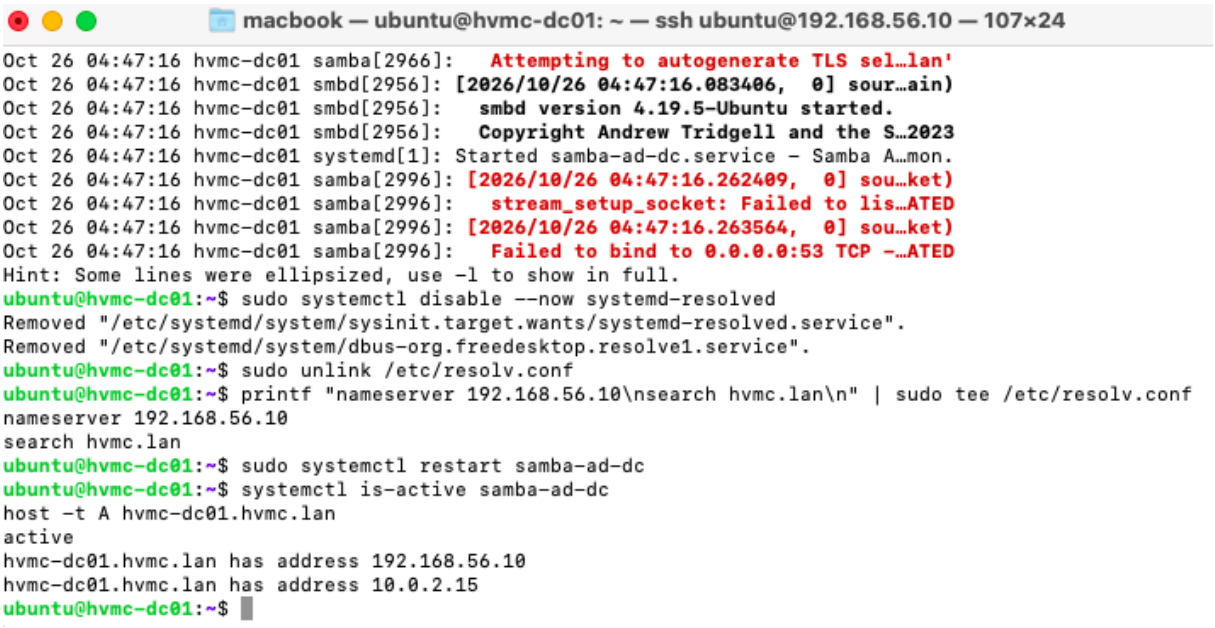


Figure 21: Applying the fix: disable systemd-resolved, new resolv.conf, restart, verify

The DNS lookup still returned a second address (10.0.2.15, the NAT adapter). Samba was restricted to the lab interface in smb.conf using `interfaces = lo enp0s8` and `bind interfaces only = yes`, and the stray DNS record was removed (it had already disappeared, so the delete reported "does not exist"). The name now resolves only to 192.168.56.10.

```
ubuntu@hvmc-dc01:~$ sudo sed -i '/^\[global\]/a interfaces = lo enp0s8\nbind interfaces only = yes' /etc/samba/smb.conf
cat /etc/samba/smb.conf
sudo systemctl restart samba-ad-dc
# Global parameters
[global]
interfaces = lo enp0s8
bind interfaces only = yes
        dns forwarder = 8.8.8.8
        netbios name = HVMC-DC01
        realm = HVMC.LAN
        server role = active directory domain controller
        workgroup = HVMC
        idmap_ldb:use rfc2307 = yes

[sysvol]
        path = /var/lib/samba/sysvol
        read only = No

[netlogon]
        path = /var/lib/samba/sysvol/hvmc.lan/scripts
        read only = No
ubuntu@hvmc-dc01:~$ sudo systemctl restart samba-ad-dc
ubuntu@hvmc-dc01:~$ samba-tool dns delete 127.0.0.1 hvmc.lan hvmc-dc01 A 10.0.2.15 -U administrator
WARNING: Using passwords on command line is insecure. Installing the setproctitle python module will hide these from shortly after program start.
Password for [HVMC\administrator]:
ERROR(runtime): The record does not exist [WERR_DNS_ERROR_RECORD_DOES_NOT_EXIST] - (9701, 'WERR_DNS_ERROR_RECORD_DOES_NOT_EXIST')
ubuntu@hvmc-dc01:~$ host -t A hvmc-dc01.hvmc.lan
hvmc-dc01.hvmc.lan has address 192.168.56.10
ubuntu@hvmc-dc01:~$
```

Figure 22: Final smb.conf and DNS check: hvmc-dc01.hvmc.lan resolves to 192.168.56.10 only

7.5 Testing authentication

A Kerberos ticket was requested for the domain administrator with `kinit` and confirmed with `klist`. A valid ticket proves that DNS, time, Kerberos and the directory are all working together. Because the lab administrator account would otherwise expire in 41 days, its password expiry was disabled with `samba-tool user setexpiry --noexpiry` (run with sudo; the first attempt without sudo was correctly refused with “Permission denied”).

```
ERROR(runtime): The record does not exist [WERR_DNS_ERROR_RECORD_DOES_NOT_EXIST] - (9701, 'WERR_DNS_ERROR_RECORD_DOES_NOT_EXIST')
ubuntu@hvmc-dc01:~$ host -t A hvmc-dc01.hvmc.lan
hvmc-dc01.hvmc.lan has address 192.168.56.10
ubuntu@hvmc-dc01:~$ kinit administrator@HVMC.LAN
Password for administrator@HVMC.LAN:
Warning: Your password will expire in 41 days on Mon 07 Dec 2026 04:32:45 AM UTC
ubuntu@hvmc-dc01:~$ klist
Ticket cache: FILE:/tmp/krb5cc_1000
Default principal: administrator@HVMC.LAN

Valid starting       Expires              Service principal
10/26/2026 05:13:45  10/26/2026 15:13:45  krbtgt/HVMC.LAN@HVMC.LAN
        renew until 10/27/2026 05:13:38
ubuntu@hvmc-dc01:~$
```

Figure 23: Kerberos ticket obtained for administrator@HVMC.LAN

```
        renew until 10/27/2026 05:13:38
ubuntu@hvmc-dc01:~$ samba-tool user setexpiry administrator --noexpiry
ltdb: tdb(/var/lib/samba/private/sam.ldb): tdb_open_ex: could not open file /var/lib/samba/private/sam.ldb: Permission denied

Unable to open tdb '/var/lib/samba/private/sam.ldb': Permission denied
Failed to connect to 'tdb:///var/lib/samba/private/sam.ldb' with backend 'tdb': Unable to open tdb '/var/lib/samba/private/sam.ldb': Permission denied
ERROR(ldb): uncaught exception - Unable to open tdb '/var/lib/samba/private/sam.ldb': Permission denied
ubuntu@hvmc-dc01:~$ sudo samba-tool user setexpiry administrator --noexpiry
Expiry for user 'administrator' disabled.
ubuntu@hvmc-dc01:~$
```

Figure 24: Password expiry disabled for the lab administrator (first attempt without sudo denied)

Snapshot 02 – “Samba domain HVMC.LAN working” was taken after a clean shutdown.

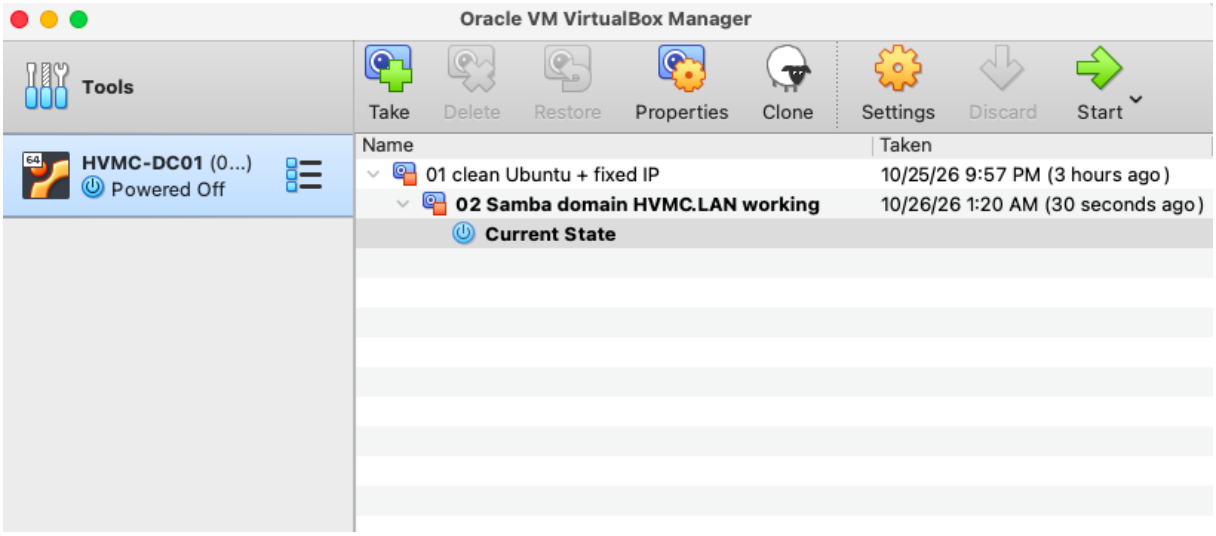


Figure 25: VirtualBox snapshots 01 and 02

8. Hospital Directory Structure

The directory was organised using the design from the Week 1 Day 2 deliverable: a top-level OU structure, one OU per department, role-based security groups named **SG-<Department>-<Application>-<Access>**, and the 30 employees from the Day 1 workforce list.

8.1 Organizational Units

Top-level OU	Purpose
Users	Employee accounts, with one sub-OU per department
Groups	Role-based security groups
Computers	Workstations joined to the domain
Service Accounts	Non-human accounts for applications
Disabled Users	Accounts of people who have left (kept for audit)

Ten department OUs were created under Users: Doctors, Nursing, Pharmacy, Laboratory, Human Resources, Finance, IT, Security, Reception, Contractors.

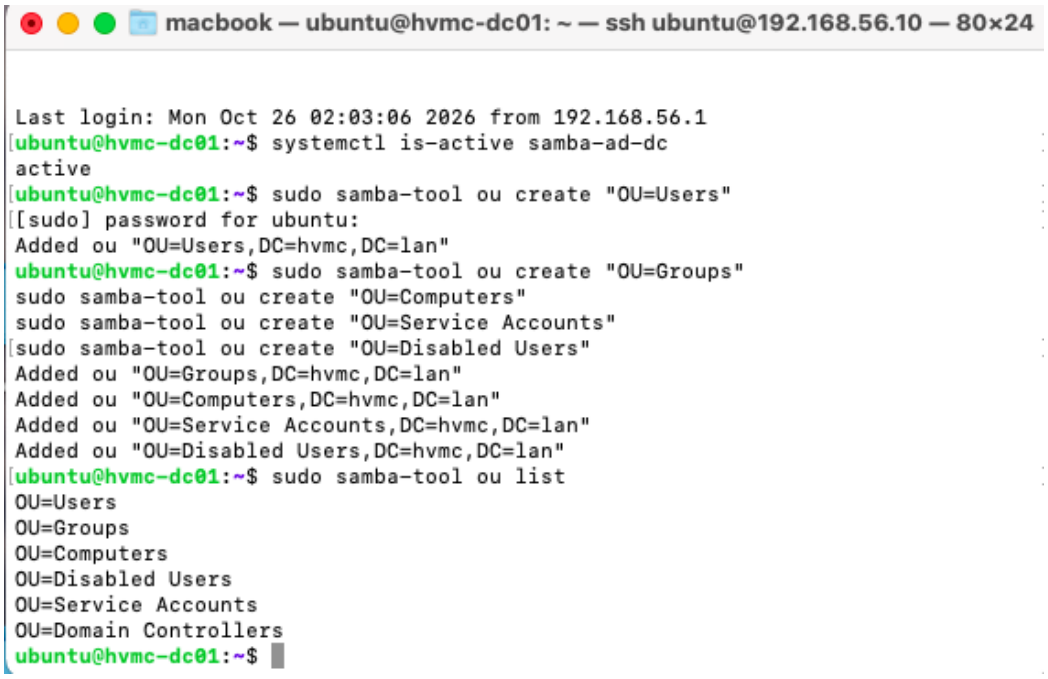


Figure 26: Top-level OUs created



Figure 27: Department OUs created under OU=Users

8.2 Security Groups

Group	Access
SG-Doctors-EHR-RW	Doctors: read/write electronic health records
SG-Nursing-EHR-RW	Nurses: read/write EHR
SG-Pharmacy-PharmacySystem-RW	Pharmacy application
SG-Lab-LIS-RW	Laboratory information system
SG-HR-HRPortal-RW	HR portal
SG-Finance-Payroll-RW	Payroll
SG-IT-ITAdmin-Admin	IT administration
SG-Security-SecurityPortal-Admin	Security monitoring
SG-Reception-Scheduling-RW	Appointment scheduling
SG-Contractors-VPN-Read	Contractor VPN, read only

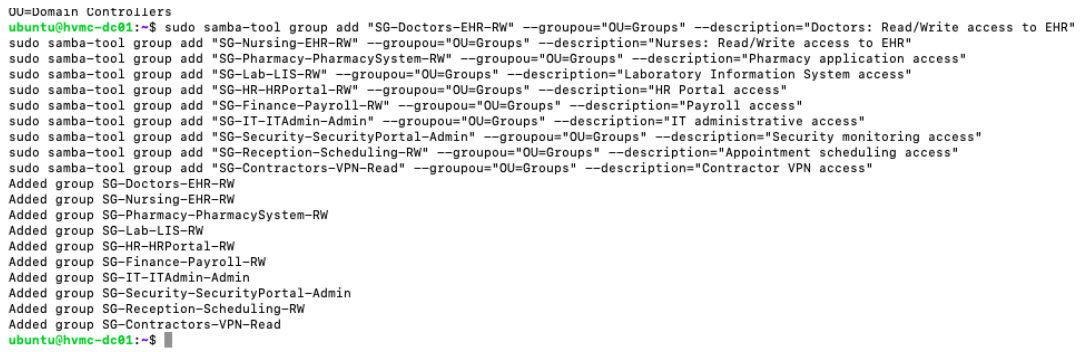


Figure 28: Ten security groups created in OU=Groups

8.3 Users and Group Membership

The 30 employee accounts were created with a script (one line per employee: username, name, title, department, group). Using a script keeps the build repeatable and avoids typing errors. All accounts use a shared practice password for the lab only. Each user was placed in the correct department OU and added to the matching security group.

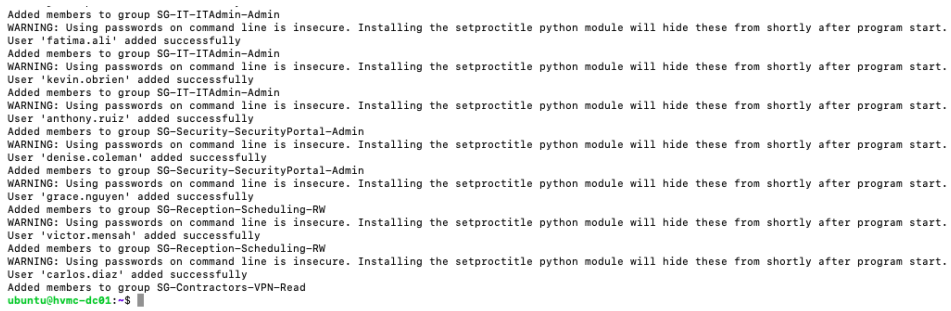


Figure 29: Users created and added to their security groups

Department	Users	Group
Doctors	5	SG-Doctors-EHR-RW
Nursing	7	SG-Nursing-EHR-RW
Pharmacy	3	SG-Pharmacy-PharmacySystem-RW
Laboratory	3	SG-Lab-LIS-RW
Human Resources	2	SG-HR-HRPortal-RW
Finance	2	SG-Finance-Payroll-RW
IT	3	SG-IT-ITAdmin-Admin
Security	2	SG-Security-SecurityPortal-Admin
Reception	2	SG-Reception-Scheduling-RW
Contractors	1	SG-Contractors-VPN-Read

```
Added members to group SG-Contractors-VPN-Read
ubuntu@hvmc-dc01:~$ for g in $(sudo samba-tool group list | grep '^SG-'); do echo "$g: $(sudo samba-tool group listmembers "$g" | wc -l)"; done
SG-Doctors-EHR-RW: 5
SG-Reception-Scheduling-RW: 2
SG-IT-ITAdmin-Admin: 3
SG-Nursing-EHR-RW: 7
SG-HR-HRPortal-RW: 2
SG-Pharmacy-PharmacySystem-RW: 3
SG-Lab-LIS-RW: 3
SG-Contractors-VPN-Read: 1
SG-Security-SecurityPortal-Admin: 2
SG-Finance-Payroll-RW: 2
ubuntu@hvmc-dc01:~$ sudo samba-tool user list | wc -l
33
ubuntu@hvmc-dc01:~$
```

Figure 30: Verification: member count per group and 33 total accounts (30 employees + built-in accounts)

9. Snapshots

Snapshots are saved restore points. Each was taken with the server powered off, so it is clean.

Snapshot	State captured
01 clean Ubuntu + fixed IP	Ubuntu installed, fixed IP 192.168.56.10, no directory software
02 Samba domain HVMC.LAN working	Domain provisioned, DNS fixed, Kerberos login tested
03 HVMC OUs, groups, 30 users	Complete hospital directory structure

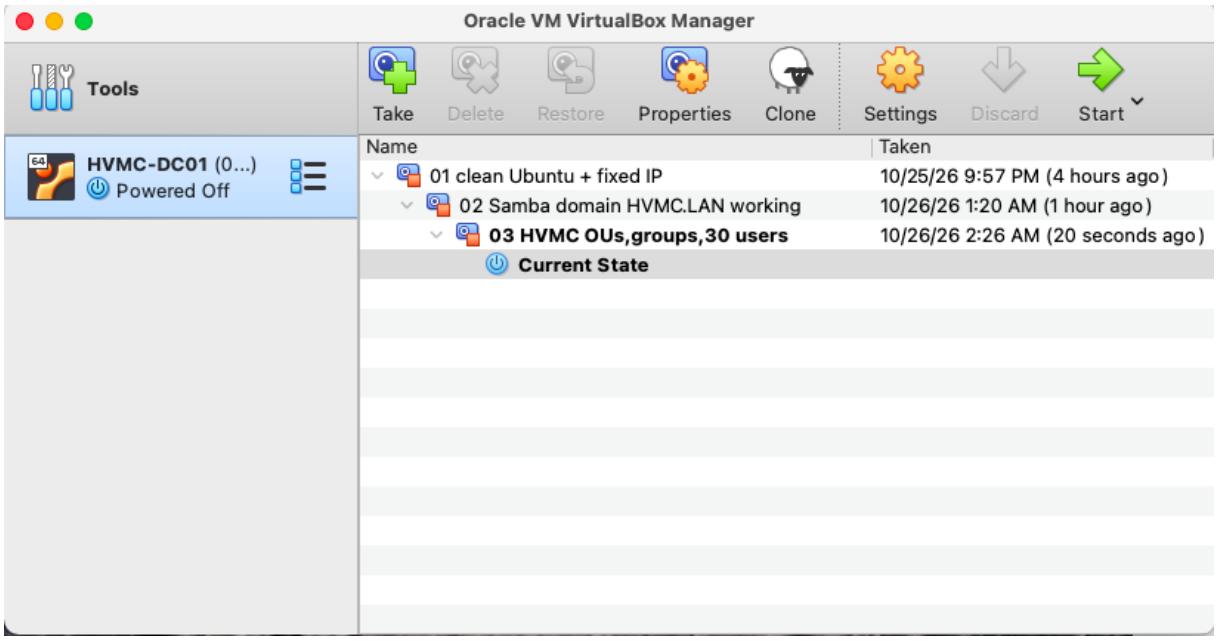


Figure 31: VirtualBox snapshot tree: 01, 02 and 03

10. Troubleshooting Log

Issue	Cause	Resolution
Samba failed to bind port 53	systemd-resolved already used DNS port	Disabled systemd-resolved; new resolv.conf; restarted samba-ad-dc
DNS returned two addresses	Samba also listened on the NAT adapter	Restricted interfaces to lo and enp0s8; bind interfaces only
setexpiry "Permission denied"	sam.ldb needs root access	Re-ran with sudo
Long script paste garbled	Terminal paste size limit	Rebuilt the script in smaller pieces; checked with bash -n and a line count of 30

11. Security Considerations

- All names, departments and users are **fictional**; no real patient or employee data is used.
- The passwords shown or used in this lab are **practice-only** credentials for an isolated, host-only network and are not used anywhere else.
- Samba listens only on the private lab interface, so the domain controller is not exposed to other networks.
- Role-based groups follow least privilege: each department receives only the access its job requires.
- Snapshots provide a quick rollback if a configuration change goes wrong.

12. Key Outcomes

- Built a working Active Directory domain controller (HVMC.LAN) on a 2014 laptop using free software.
- Configured fixed addressing, DNS and Kerberos, and diagnosed and fixed a real DNS port conflict.
- Created 5 top-level OUs, 10 department OUs, 10 role-based security groups and 30 users using repeatable command-line tooling.
- Protected the work with three verified snapshots.

Next steps: build the Windows-style client workstation HVMC-CLIENT01 (192.168.56.11), join it to HVMC.LAN and sign in as a domain user.

13. Deliverables

- Week 1 – Virtual Machine & Domain Controller Build: Lab Documentation for GitHub Portfolio (this document).
- VM snapshots 01–03 of HVMC-DC01 (kept locally).